



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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Concentration Inequalities (AI2200)

Lecture 06

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Sub-Gaussian Random Variables

Theorem 1 (Equivalent Statements for sub-Gaussian Random Variables)

The following statements are equivalent.

1. There exists a constant $K_1 > 0$ such that $\mathbb{P}(|X| \geq \varepsilon) \leq 2e^{-\varepsilon^2/K_1^2} \quad \forall \varepsilon > 0.$
2. There exists a constant $K_2 > 0$ such that $(\mathbb{E}[|X|^p])^{1/p} \leq K_2 \sqrt{p} \quad \forall p \geq 1.$
3. There exists a constant $K_3 > 0$ such that $\mathbb{E}[e^{t^2 X^2}] \leq e^{t^2 K_3^2} \quad \forall t \in [-\frac{1}{K_3}, \frac{1}{K_3}].$
4. There exists a constant $K_4 > 0$ such that $\mathbb{E}[e^{X^2/K_4^2}] \leq 2.$
5. (Applicable if $\mathbb{E}[X] = 0$)
There exists a constant $K_5 > 0$ such that $\mathbb{E}[e^{tX}] \leq e^{K_5^2 t^2} \quad \forall t \in \mathbb{R}.$

Notes:

- If any of the above statements holds for some constant K , it also holds for a larger constant than K
- If $K_1^*, \dots, K_5^*$ denote the best constants, then for any $i \neq j$, there exist universal constants c_{ij}, C_{ij} such that

$$c_{ij} \leq K_i^*/K_j^* \leq C_{ij}.$$

Here, c_{ij} and C_{ij} do not depend on the distribution of X

A Corollary to Theorem 1

- If X is a bounded random variable, i.e., there exists $M > 0$ such that

$$\mathbb{P}(|X| \leq M) = 1,$$

then X is sub-Gaussian

Proof of corollary:

- For any $p \geq 1$, we note that

$$\mathbb{P}(|X|^p \leq M^p) = 1, \quad \mathbb{E}[|X|^p] \leq M^p \leq M^p p^{p/2}.$$

- We therefore have

$$\left(\mathbb{E}[|X|^p]\right)^{1/p} \leq M\sqrt{p} \quad \forall p \geq 1.$$

Sub-Gaussian Norm

Definition (Sub-Gaussian Norm)

For a sub-Gaussian random variable X , its **sub-Gaussian norm** is defined as

$$\|X\|_{\psi_2} := \inf \left\{ K > 0 : \mathbb{E} \left[e^{X^2/K^2} \right] \leq 2 \right\}.$$

- If X is sub-Gaussian, then αX is sub-Gaussian for all $\alpha \in \mathbb{R}$. Furthermore,

$$\|\alpha X\|_{\psi_2} = |\alpha| \|X\|_{\psi_2}.$$

- If X, Y are sub-Gaussian with sub-Gaussian norms $K_X = \|X\|_{\psi_2}$ and $K_Y = \|Y\|_{\psi_2}$ respectively, then

$$\|X + Y\|_{\psi_2} \leq \|X\|_{\psi_2} + \|Y\|_{\psi_2} = K_X + K_Y.$$

- More generally, for any $a_1, \dots, a_n \in \mathbb{R}$ and sub-Gaussian random variables $X_1, \dots, X_n$,

$$\left\| \sum_{k=1}^n a_k X_k \right\|_{\psi_2} \leq \sum_{k=1}^n |a_k| \|X_k\|_{\psi_2}.$$

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- If X, Y are **independent** and sub-Gaussian with $\|X\|_{\psi_2} = K_X$ and $\|Y\|_{\psi_2} = K_Y$ respectively, then

$$\|X + Y\|_{\psi_2} \leq \sqrt{\|X\|_{\psi_2}^2 + \|Y\|_{\psi_2}^2} = \sqrt{K_X^2 + K_Y^2}.$$

Proof:

- Without loss of generality, let $\mathbb{E}[X] = 0 = \mathbb{E}[Y]$
- Using Property 5 of Theorem 1, we have that there exists a universal constant $c > 0$ such that

$$\forall t \in \mathbb{R}, \quad \mathbb{E}[e^{tX}] \leq e^{ct^2 K_X^2}, \quad \mathbb{E}[e^{tY}] \leq e^{ct^2 K_Y^2}.$$

- Then, we have

$$\mathbb{E} \left[e^{t(X+Y)} \right] \leq e^{ct^2 (K_X^2 + K_Y^2)},$$

from which it follows that

$$\|X + Y\|_{\psi_2}^2 \leq K_X^2 + K_Y^2.$$

Hoeffding's Inequality

Theorem 2 (Hoeffding's Inequality)

Suppose that $X_1, \dots, X_n$ are independent and sub-Gaussian with sub-Gaussian norms $K_1, \dots, K_n$ respectively.

Let $\mu_1, \dots, \mu_n$ denote the means of $X_1, \dots, X_n$ respectively.

Then, for any $\alpha_1, \dots, \alpha_n \in \mathbb{R}$,

$$\mathbb{P} \left(\left\{ \left| \sum_{\ell=1}^n \alpha_{\ell} (X_{\ell} - \mu_{\ell}) \right| > \varepsilon \right\} \right) \leq 2 \exp \left(- \frac{c \varepsilon^2}{\sum_{\ell=1}^n \alpha_{\ell}^2 \left(K_{\ell}^2 + \frac{\mu_{\ell}^2}{\log 2} \right)} \right) \quad \forall \varepsilon > 0,$$

where c is a universal constant that does not depend on the distributions of $X_1, \dots, X_n$.

Remark:

- Using the relation (this property will be proved formally in the next lecture) that

$$\|X_{\ell} - \mu_{\ell}\|_{\psi_2}^2 \leq 2 \|X_{\ell}\|_{\psi_2}^2 \quad \forall \ell,$$

one can derive a version of the Hoeffding's inequality without the $\frac{\mu_{\ell}^2}{\log 2}$ term in the denominator

Proof of Theorem 2

- We will apply Theorem 1 to the random variable

$$X = \sum_{\ell=1}^n \alpha_{\ell} (X_{\ell} - \mu_{\ell}).$$

- For each $\ell \in \{1, \dots, n\}$, we have

$$\|\alpha_{\ell} (X_{\ell} - \mu_{\ell})\|_{\psi_2}^2 \leq \alpha_{\ell}^2 (K_{\ell}^2 + \mu_{\ell}^2).$$

- As a consequence, we have

$$\mathbb{E} \left[\exp \left(t \sum_{\ell=1}^n \alpha_{\ell} (X_{\ell} - \mu_{\ell}) \right) \right] = \prod_{\ell=1}^n \mathbb{E} [e^{t \alpha_{\ell} (X_{\ell} - \mu_{\ell})}] \stackrel{\text{P5, Th.1}}{\leq} \prod_{\ell=1}^n e^{t^2 \alpha_{\ell}^2 \| \alpha_{\ell} (X_{\ell} - \mu_{\ell}) \|_{\psi_2}^2} \leq \prod_{\ell=1}^n e^{t^2 \alpha_{\ell}^2 \left(K_{\ell}^2 + \frac{\mu_{\ell}^2}{2} \right)}$$

- Using the equivalence between parts 1, 5 of Theorem 1 for the random variable X , we arrive at the desired result

Application: Multi-Armed Bandits

- Suppose that there are K slot machines, each generating sub-Gaussian rewards
- Let the reward distribution of slot machine k be denoted by ν_k
- At each round $t \in \{1, 2, \dots\}$, you may play any one of the slot machines
- Let A_t be the slot machine played at time t , and let R_t be the corresponding reward generated
- By playing for a total of T rounds, the goal is to maximise the expected total reward:

$$\max \mathbb{E} \left[\sum_{t=1}^T R_t \right] .$$